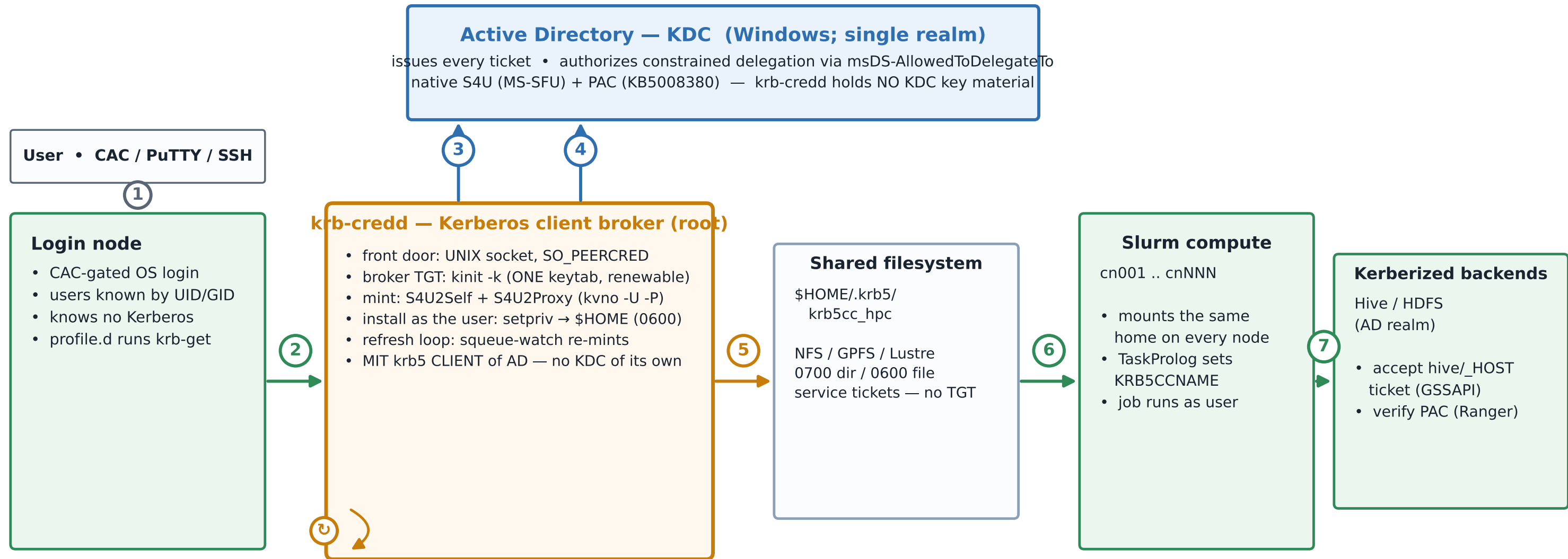


krb-credd — Kerberos Client Broker (root): System Architecture

One broker keytab → S4U service tickets from Active Directory → installed as the user into shared-home → visible on every Slurm compute node



End-to-end flow

- 1** CAC / PuTTY / SSH login establishes the user's UID/GID on the login node (no Kerberos on the HPC side).
 - 2** profile.d runs krb-get → krb-credd over a UNIX socket; the UID comes from SO_PEERCREC (kernel-verified, never the request).
 - 3** krb-credd keeps its broker TGT fresh: kinit -k from ONE keytab (renewable, unattended)—obtained from AD.
 - 4** krb-credd mints the user's SERVICE tickets: S4U2Self + S4U2Proxy (kvno -U user -P spns); AD authorizes via msDS-AllowedToDelegateTo.
 - 5** krb-credd installs the cache AS THE USER (setpriv, atomic, 0700/0600) into \$HOME/.krb5/krb5cc_hpc on shared storage.
 - 6** Every compute node mounts the same home; the Slurm TaskProlog points KRB5CCNAME at the cache — no copy, no node daemon.
 - 7** The job uses the hive/_HOST (etc.) service ticket over GSSAPI; the backend verifies it and checks the PAC (Ranger).
- 🔄** The squeue-watch refresh loop RE-MINTS service tickets before expiry, so long or queued jobs never find them stale.